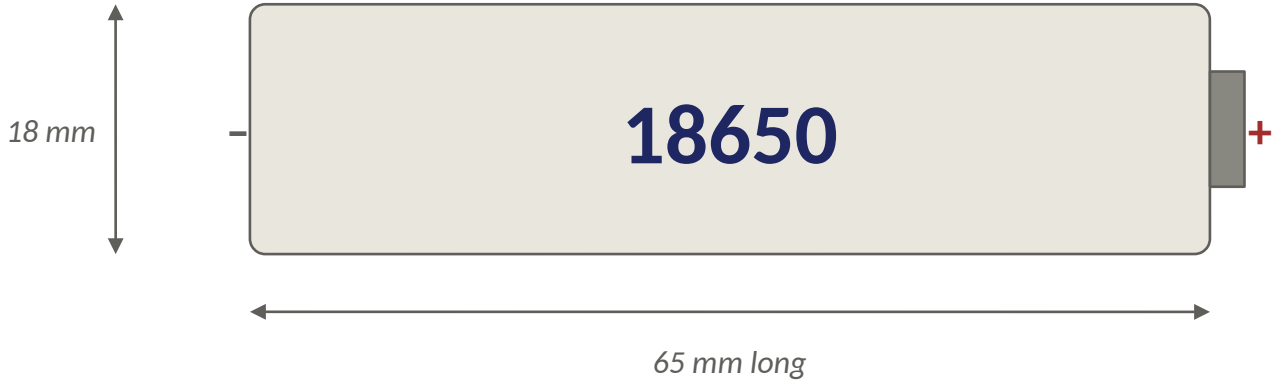
Batteries & Power Electronics

A robotics class review

Tool batteries · FIRST robotics · transformers · rectifiers · buck converters · USB PD

The 18650 cell

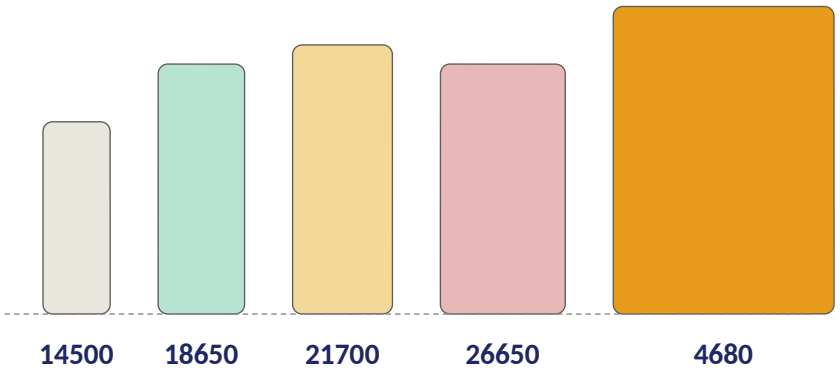
The workhorse cylindrical lithium-ion cell — and where its name comes from



"18" = diameter (mm) **"65"** = length (mm) **"0"** = cylindrical shape

Cylindrical cell sizes

Tools and EVs are moving to bigger cells — more energy per cell, fewer welds per pack

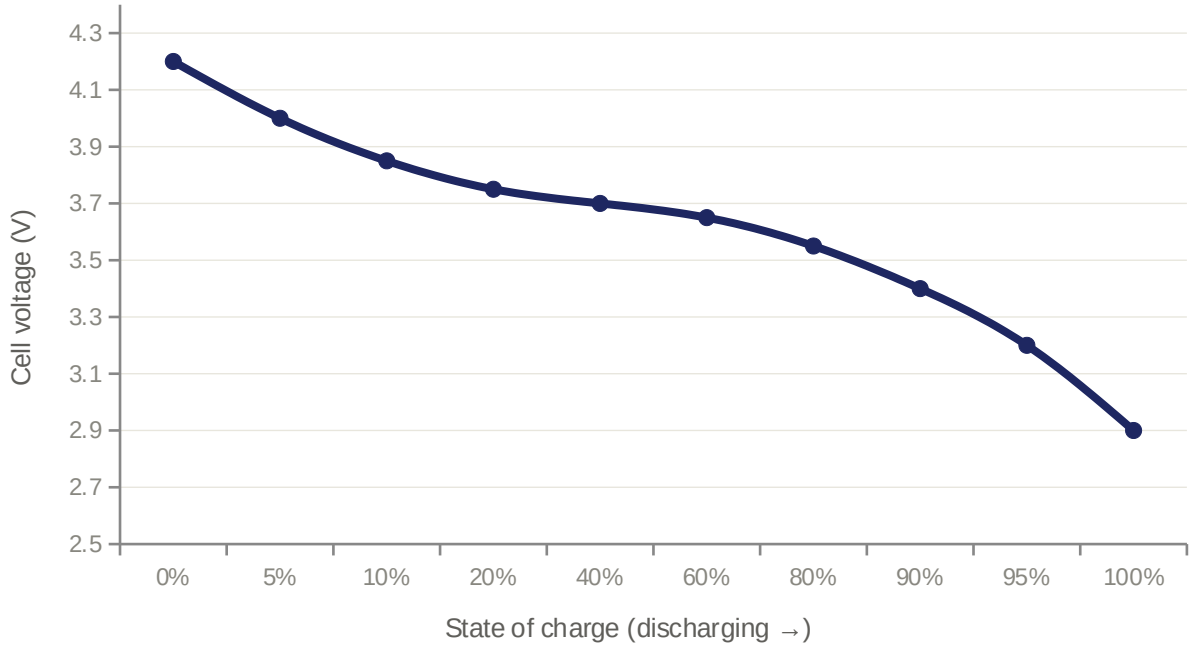


relative size

Cell	Where you'll find it
14500	AA-sized lithium · vapes, small flashlights
18650	Old laptops, flashlights, original Tesla Roadster
21700	Modern tool packs, Tesla Model 3/Y, Rivian
26650	High-current flashlights, e-bikes, industrial
4680	Tesla Model Y (new), Cybertruck, BMW Neue Klasse

Nominal voltage vs peak charge

Why a lithium cell doesn't have just one voltage



4.2 V peak

just after charging

→ 5 cells × 4.2 = 21 V

3.6 V nominal

average during use

→ 5 cells × 3.6 = 18 V

3.0 V cutoff

BMS shuts off discharge

The 18V / 20V mystery

Same five cells. Same chemistry. Just different numbers on the label.

Labeled as NOMINAL

18V

5 cells × 3.6V = 18V

Milwaukee M18
Bosch 18V
Makita LXT

=

Labeled as PEAK

20V Max

5 cells × 4.2V ≈ 21V (rounded)

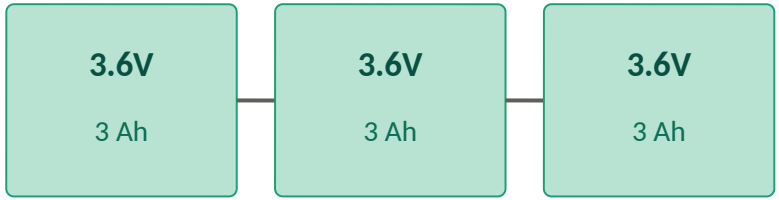
DeWalt
Black & Decker
Porter-Cable

Marketing, not engineering — both labels point to the same physical pack at different points on the discharge curve

Series vs parallel

The foundation of every battery pack

SERIES



3 × 3.6V = 10.8 V total

voltage adds · capacity stays at 3 Ah

PARALLEL



3.6 V × 9 Ah

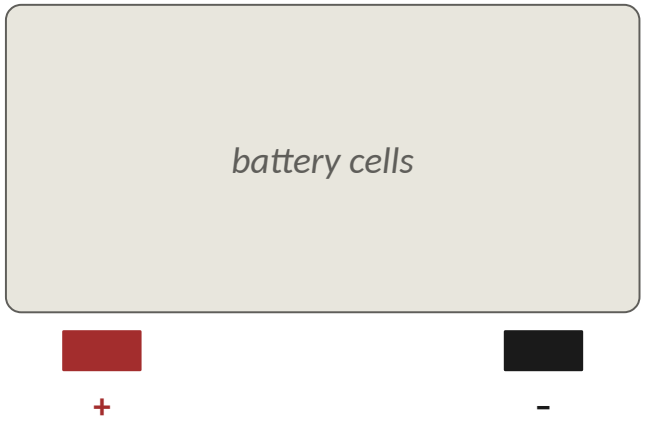
capacity adds · voltage stays at 3.6 V

"5S2P" = 5 series + 2 parallel = 10 cells total. Combines both to get the voltage AND capacity you need.

High-performance batteries

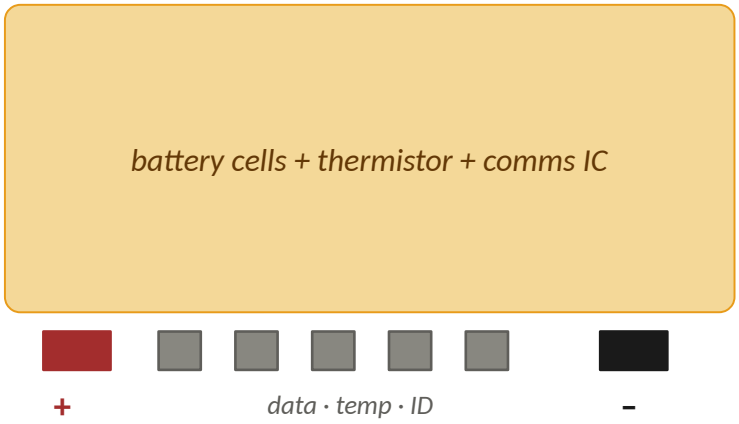
Why brushless tool packs have extra contacts

STANDARD PACK



two contacts · power only

HIGH-PERFORMANCE PACK



multiple contacts · pack and tool talk to each other

Tool reads pack temp, state of charge, cell health, and pack ID — throttles or refuses to fire if anything's wrong

Lithium cell chemistries

The cathode material changes everything — energy, current, safety, cost

Chemistry	Nominal V	Energy (Wh/kg)	Cycles	Best for
NMC	3.6V	150–220	1k–2k	Modern EVs, quality tool packs · balanced everything
LFP	3.2V	90–160	2k–6k+	Safest · solar storage, Tesla SR · long life, no cobalt
LMO / IMR	3.7V	100–150	300–700	Tools, vapes · high current bursts
LCO / ICR	3.6V	150–200	500–1k	Phones, laptops · high capacity, low current
NCA	3.6V	200–260	500–1.5k	Tesla Model S/X, aerospace · highest energy density
LTO	2.4V	50–80	10k–30k	Grid storage, fast-charge buses · ultra-long life

Quick tag decode:* *IMR *= high current (manganese), used in tools and vapes.* ***ICR*** *= high capacity (cobalt), used in laptops.*

Pouch cells vs cylindrical

Trade rigid metal for a flexible foil envelope

POUCH



flat · flexible foil

- Higher energy density per volume
- Shape-flexible — fits any device
- Lighter — no steel shell
- Easy to puncture
- Swells with age
- Custom per product

CYLINDRICAL (18650 / 21700)

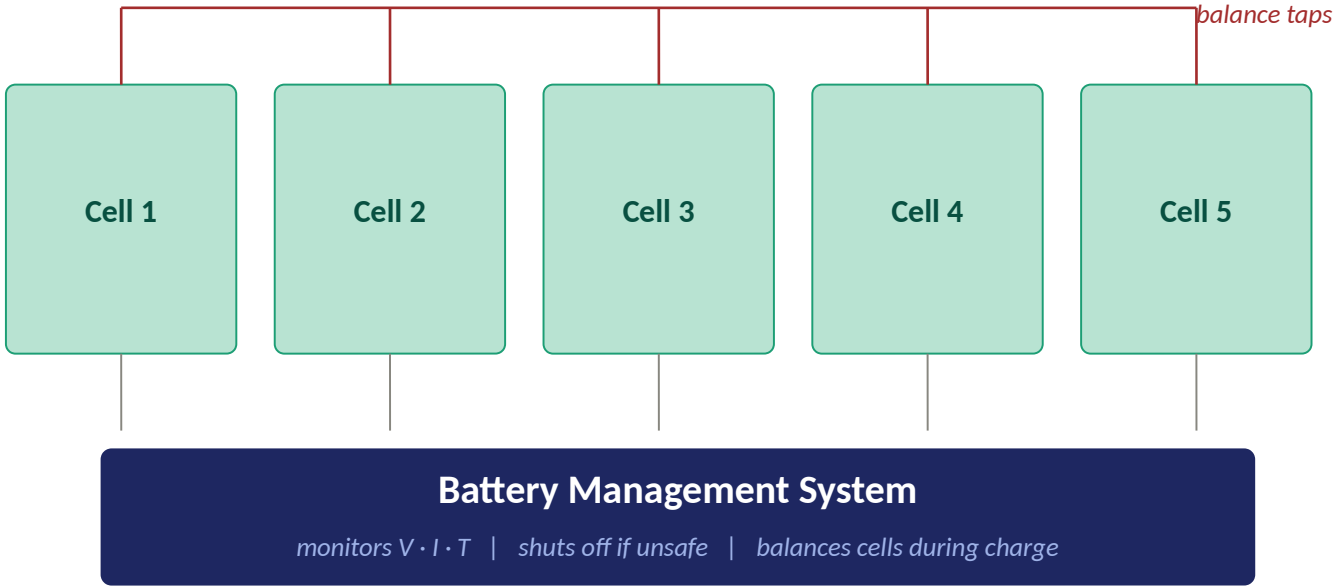


round · steel can

- Standardized · universally available
- Self-supporting · survives drops
- Easier thermal management
- Air gaps waste pack space
- Heavier per Wh (steel)
- Fixed sizes only

Battery Management System (BMS)

Watches every cell individually · balances during charge · cuts off if unsafe



Never build a multi-cell lithium pack without a BMS. One cell drifting low or high is a fire risk.

FIRST Robotics uses AGM, not lithium

Sealed Lead-Acid (SLA) family — and why AGM is the right pick

SLA family tree

Flooded Lead-Acid

Original. Liquid electrolyte. Spills, vents hydrogen.

Gel Cell

Silica-thickened. Spill-proof but slow charge only.

AGM — Absorbed Glass Mat

Electrolyte soaked into a fiberglass mat between plates.

High surge current · spill-proof · any orientation · what FRC uses

Key specs

Nominal voltage	**2.0 V / cell**
Full charge	**2.25–2.30 V**
Cutoff	**1.75 V / cell**
Energy density	**30–50 Wh/kg**
Cycle life	**200–500**
Self-discharge	**3–5% / mo**

Why AGM is right for FIRST

Specifically suited to competition robots and student teams

1 Brutal surge current

Drivetrain motors pull 100+ amps in a stall. AGM handles this without complaint.

2 Safe for student use

No fire risk, no spill risk. Survives charging mistakes that would destroy lithium.

3 Any orientation

Robots flip and roll. AGM is sealed — mounts upside-down, sideways, anywhere.

4 No BMS required

Six cells, two terminals, done. Much simpler electrical system than lithium.

5 Cheap and standardized

Every team uses the same model. Easy to source, easy to replace.

6 Voltage sag = current limit

Natural drop under heavy load inherently protects the system — lithium has no such grace.

SLA / AGM vs Lithium-Ion

Head-to-head across the metrics that matter for robots

	SLA / AGM	Lithium-Ion (LFP)
Energy density	**30–50 Wh/kg**	**90–160 Wh/kg**
Cycle life	**200–500**	**2,000–6,000**
Usable capacity	**50% (DoD limit)**	**80–100%**
Cost per kWh	**$50–$150 (cheapest)**	**$200–$400**
Weight (12V / 100Ah)	**~30 kg**	**~12 kg**
Needs BMS	**No**	**Yes**
Cold performance	**Good**	**Poor (LFP)**
Voltage sag under load	*Significant*	*Flat until cutoff*

SLA wins on cost, safety, and forgiveness — exactly why FRC uses it. Lithium wins on weight and runtime — exactly why EVs use it.

AC vs DC

The basics before we get to transformers

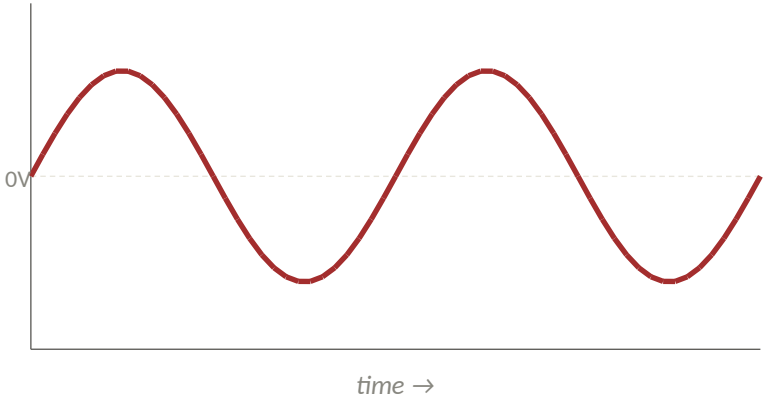
DC — Direct Current



Flat. One direction. One voltage.

Batteries · Solar panels · USB

AC — Alternating Current

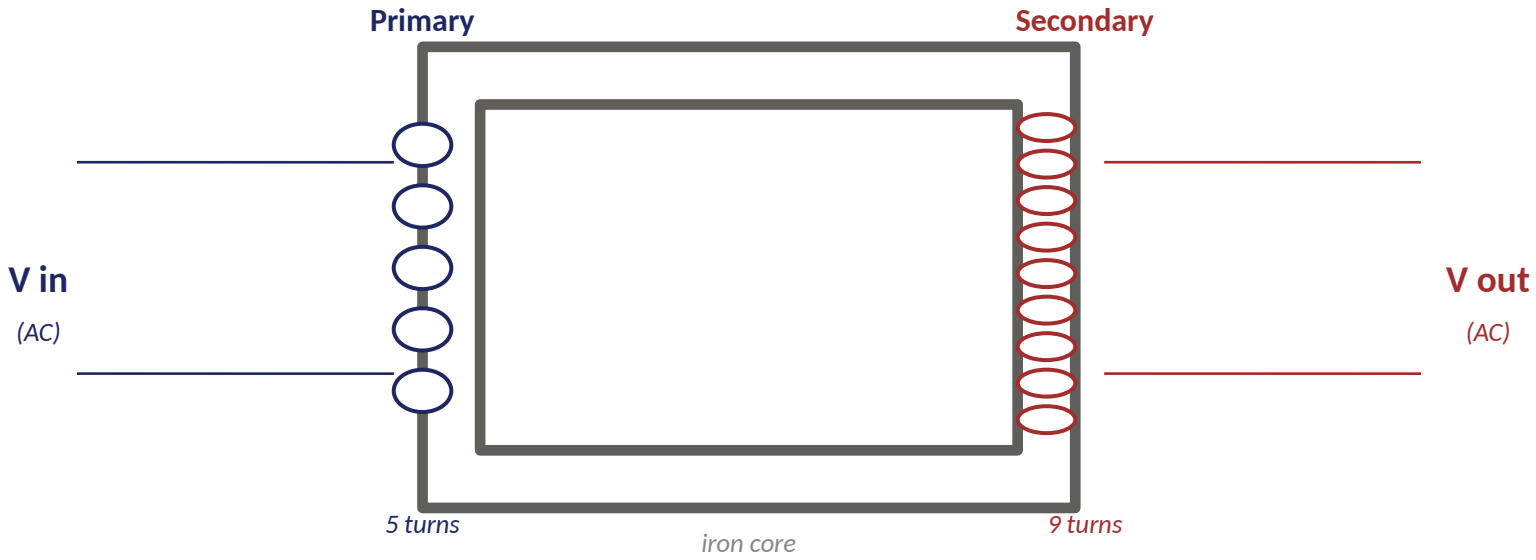


Reverses 60× per second (US wall power).

Wall outlets · Generators · Transformers

Transformers

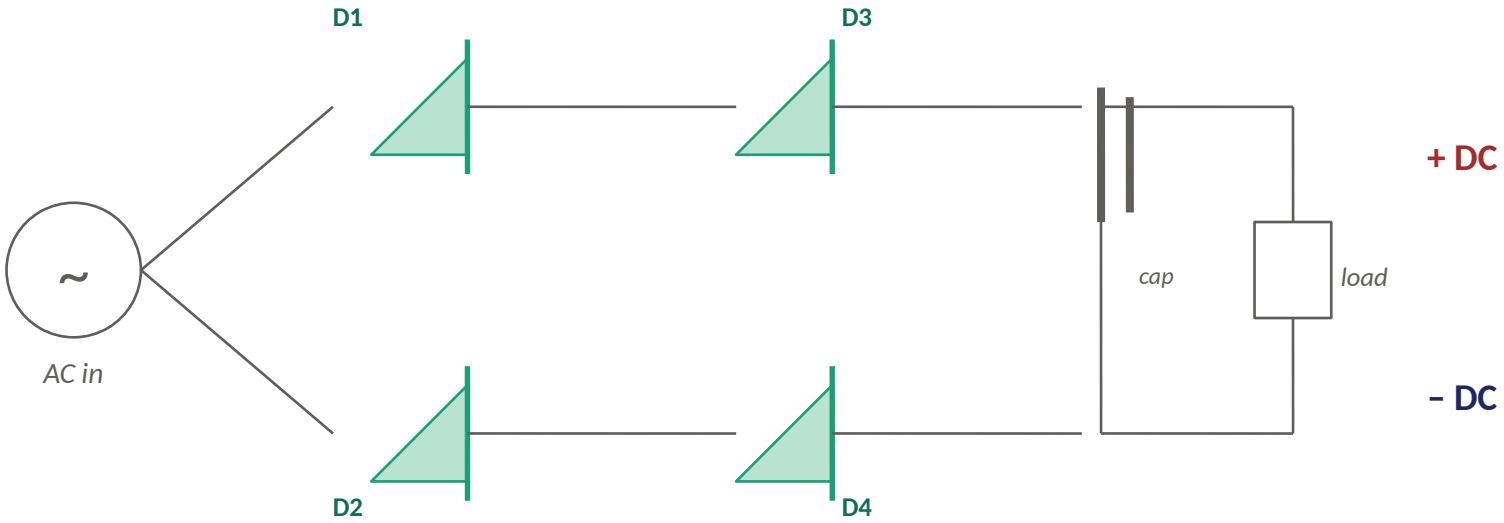
Two coils on a shared iron core — voltage ratio = turns ratio · only works with AC



9 / 5 ratio = step-up by 1.8×. Step-down = fewer secondary turns. DC won't work — needs a CHANGING magnetic field.

Full-bridge rectifier

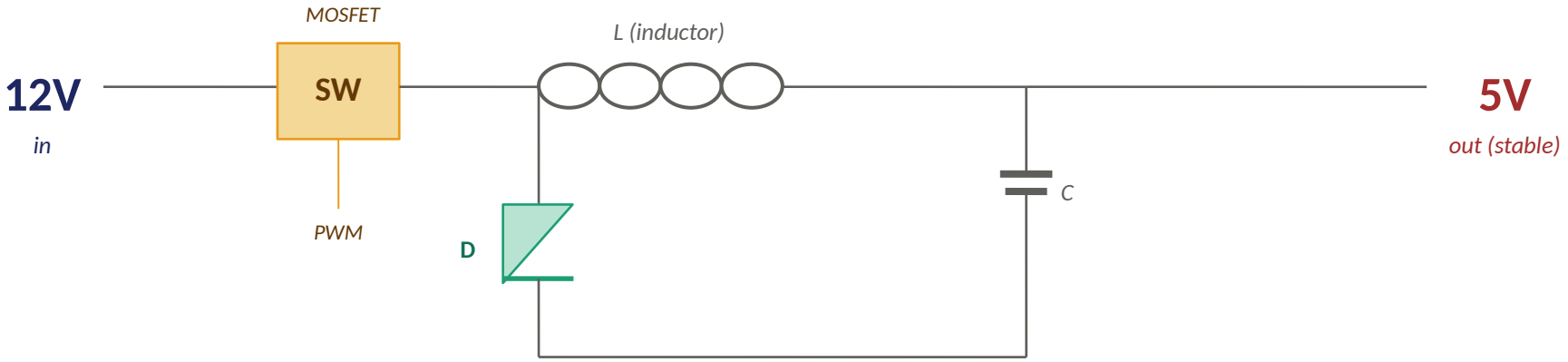
Converts AC to DC using four diodes in a diamond



Whichever way AC swings, the output stays + on top, – on bottom. Capacitor smooths the result into steady DC.

Buck converter

DC-to-DC step-down · 85–95% efficient instead of dissipating heat



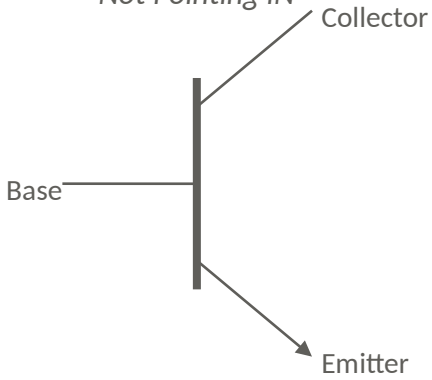
Switch chops the input · inductor smooths the current · cap smooths the voltage. Buck = down · Boost = up · Buck-boost = either.

NPN vs PNP transistors

Two complementary flavors of BJT (and same split for N-channel / P-channel MOSFET)

NPN

"Not Pointing iN"

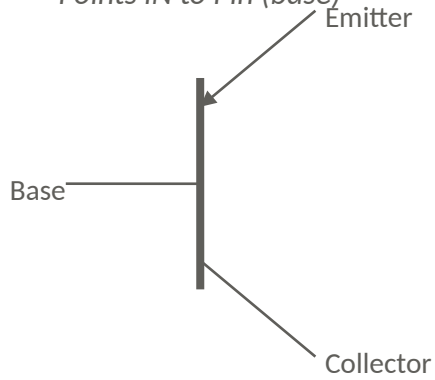


ON when base > emitter (high signal)

Used as a LOW-SIDE switch (between load and ground)

PNP

"Points iN to Pin (base)"

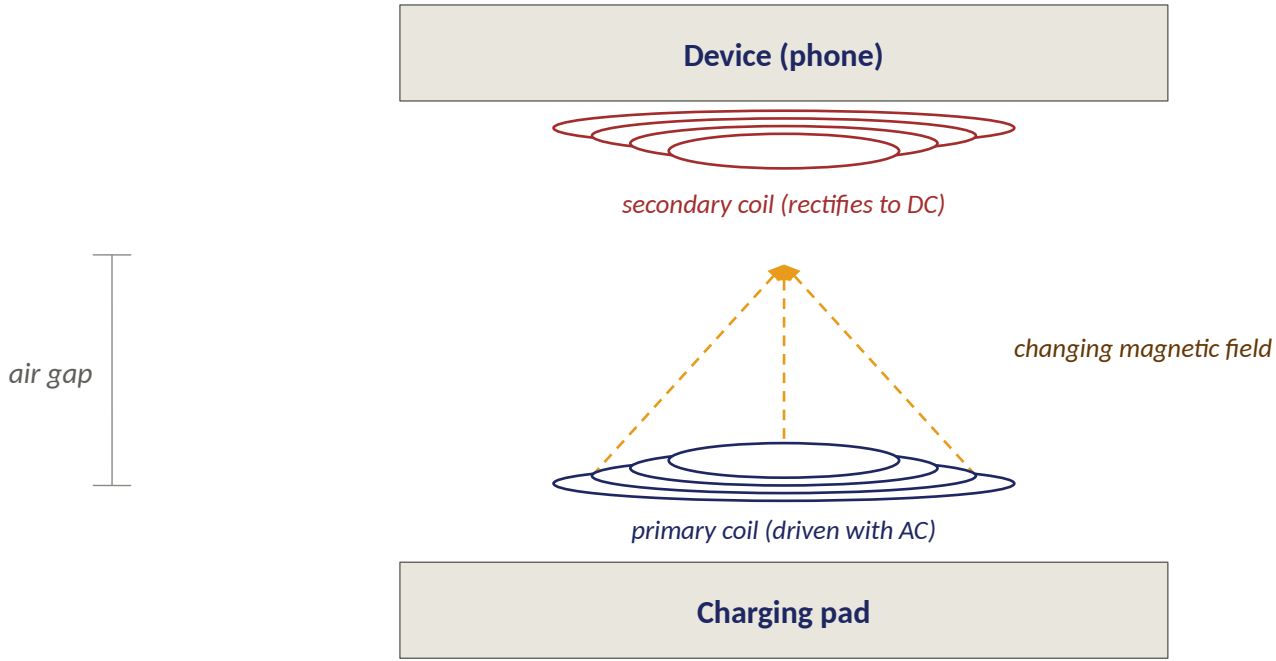


ON when base < emitter (low signal)

Used as a HIGH-SIDE switch (between supply and load)

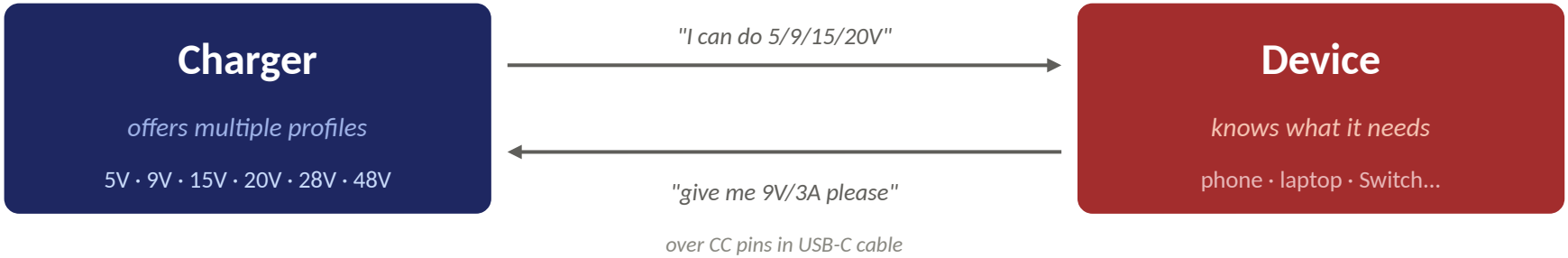
Wireless charging

It's a transformer with an air gap



USB Power Delivery (USB PD)

Charger and device negotiate the right voltage in real time

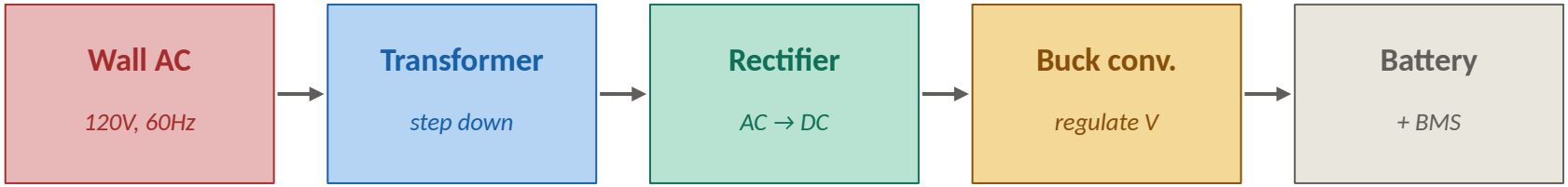


Charger reconfigures its internal buck converter to deliver the requested voltage

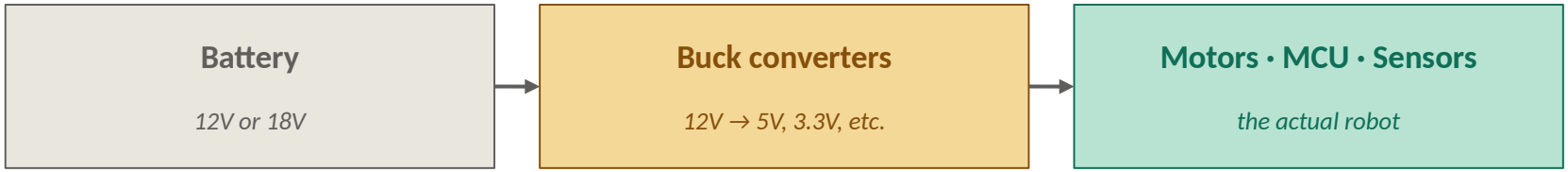
Standard	Voltage	Max power	Typical use
USB 2.0 default	5V	2.5 W	Basic accessories
USB PD 2.0/3.0	up to 20V	100 W	Laptop charging
USB PD 3.1 EPR	up to 48V	240 W	Gaming laptops, workstations

The big picture

Every battery system you'll meet uses the same stages — in some order



Powering the robot:



Get electrons into chemistry that stores them, protect them, and convert to whatever voltage the load needs.

That's the whole thing.

Cells store energy. BMS protects them.

Transformers, rectifiers, and buck converters reshape voltage.

Everything else is just packaging.

Questions? Ask the student who took this lesson.